



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2008
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Each Section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 84

- Attempt questions 1 – 6
- Board approved calculators may be used.
- Full marks may not be awarded for careless or badly arranged work.

Examiner: *A.M. Gainford*

Section A
(Start a new answer sheet.)

Question 1. (14 marks)

- | | Marks |
|--|--------------|
| (a) Given the complex number $z = -1 + \sqrt{3}i$: | 3 |
| (i) Express z in modulus-argument form. | |
| (ii) Hence evaluate z^9 . | |
|
(b) Sketch the region in the Argand diagram which simultaneously satisfies: | 3 |
| $\frac{\pi}{4} \leq \arg(z-i) \leq \frac{3\pi}{4} \quad \text{and} \quad z-i < 2.$ | |
|
(c) Find the square roots of $8-6i$. | 2 |
|
(d) Find the two complex numbers which satisfy: | 3 |
| $3z\bar{z} + 2(z - \bar{z}) = 39 + 12i$ | |
|
(e) If $z = \cos \theta + i \sin \theta$, show that $1+z = 2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$. | 3 |

Question 2. (14 marks)

Marks

- (a) Evaluate:

2

$$\int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-4x^2}} dx$$

- (b) Use the substitution $u^6 = x$ to evaluate:

3

$$\int_0^1 \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx$$

- (c) Use integration by parts to find:

3

$$\int e^x \cos x dx$$

- (d) Find:

3

$$\int \frac{dx}{x^2 - x - 6}$$

- (e) Use the substitution $t = \tan \frac{x}{2}$ to evaluate

3

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x + \sin x} dx$$

Give your answer in simplest exact form.

Section B
(Start a new answer sheet.)

Question 3. (14 marks)

Marks

- (a) The equation $x^3 + kx + 2 = 0$ has roots α , β and γ . **5**
- (i) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$, in terms of k .
- (ii) Find the monic cubic equation with roots $\alpha^2, \beta^2, \gamma^2$, in terms of k .
- (iii) Show that the value of $\alpha^3 + \beta^3 + \gamma^3$ is independent of k .
- (b) (i) Prove that if α is a double root of the polynomial equation $P(x) = 0$, **2**
then $P'(\alpha) = 0$.
- (ii) Find all the roots of the equation **3**
$$18x^3 + 3x^2 - 28x + 12 = 0$$

given that two of the roots are equal.
- (c) Given that $f(x) = \frac{11-x}{x^2-x-2}$: **4**
- (i) Graph $y = f(x)$, clearly marking its asymptotes and stationary points, if any. (Use approximate values.)
- (ii) Hence, or otherwise, graph $y = \frac{1}{f(x)}$.

Question 4 (14 marks)

- (a) Let $OABC$ be a square on the Argand diagram, where O is the origin, and C is to the left of A . The point A represents the complex number $z = 2 + 3i$. 4

- (i) Find the complex numbers represented by B and C in the form $a + ib$.
- (ii) The square is now rotated about O through 45° in the anticlockwise direction to $OA'B'C'$. Find the complex numbers represented by the points A' , B' , C' , in the form $a + ib$.

- (b) (i) Use the substitution $x = a - t$, where a is a constant, to prove that: 2

$$\int_0^a f(x) dx = \int_0^a f(a-t) dt$$

- (ii) Hence, or otherwise, show that $\int_0^2 x^2 (2-x)^9 dx = 6\frac{34}{165}$. 2

- (c) A particle moves in a straight line with velocity given by $v^2 = 36 - 4x^2$, where x is the displacement in metres from a fixed point O and t is the time in seconds. 2

- (i) Show that the motion is simple harmonic.
- (ii) Find the period of the motion.

- (d) Express the five fifth roots of unity in modulus-argument form, and clearly show them on an Argand diagram. 4

Section C
(Start a new answer booklet)

Question 5 (14 marks)

- (a) Solve for z : **3**

$$z^2 + (1 + 2i)z - (2 - 4i) = 0$$

Give your answers in the form $a + ib$.

- (b) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ where n is a positive integer. **5**

- (i) Find the value of I_1 .

- (ii) Using integration, show that $I_n + I_{n-2} = \frac{1}{n-1}$.

- (iii) Evaluate $J = \int_0^{\frac{\pi}{4}} \tan^{11} x \, dx$.

- (c) Use de Moivre's theorem to express $\cos 4\theta$ in terms of $\cos \theta$. **2**

- (d) Given that $2 + i$ is a root of the equation $x^3 - x^2 - 7x + 15 = 0$, find all roots of the equation. **4**

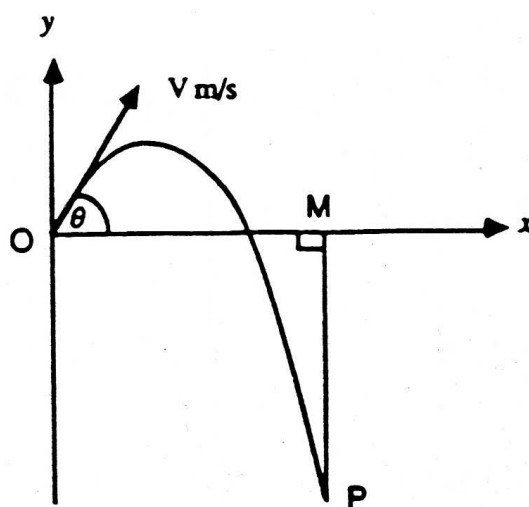
Question 6 (14 marks)

(a) Consider the function $y = e^{-x^2}$:

6

- (i) Sketch the graph of the function in the domain $-2 \leq x \leq 2$.
- (ii) Show, using your graph, that $1 < \int_{-2}^2 e^{-x^2} dx < 4$.
- (iii) Approximate $\int_{-2}^2 e^{-x^2} dx$ correct to two decimal places using Simpson's Rule with five x -values.

(c)



A projectile is fired from a point O with initial speed V m/s at an angle of elevation of θ . It is subject only to the force of gravity, resulting in a constant downward acceleration of magnitude g m/s².

8

- (i) Find functions for its horizontal (x) and vertical (y) displacements from O after t seconds.
- (ii) The projectile falls to a point P , below the level of O , such that $PM=OM$.
Prove that the time taken to reach P is $\frac{2V(\sin \theta + \cos \theta)}{g}$ seconds.
- (iii) Show that the distance OM is $\frac{V^2(\sin 2\theta + \cos 2\theta + 1)}{g}$ metres.
- (iv) If the horizontal range of the projectile level with O is r m and the distance OM is $\frac{4r}{3}$, prove that $\sin 2\theta - 3 \cos 2\theta = 3$, and find the value of $\tan \theta$.

This is the end of the paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

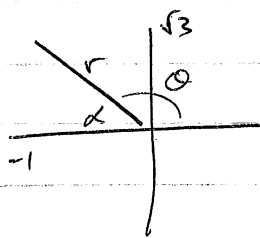
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

$$\text{NOTE: } \ln x = \log_e x, x > 0$$

Question 1

(a) i) $z = -1 + \sqrt{3}i$



$$\tan \alpha = \sqrt{3}$$

$$r^2 = (1)^2 + (\sqrt{3})^2$$

$$\alpha = \frac{\pi}{3}$$

$$r^2 = 4$$

$$\theta = \frac{2\pi}{3}$$

$$r = 2$$

$$z = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

ii) $z^9 = \left[2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^9$

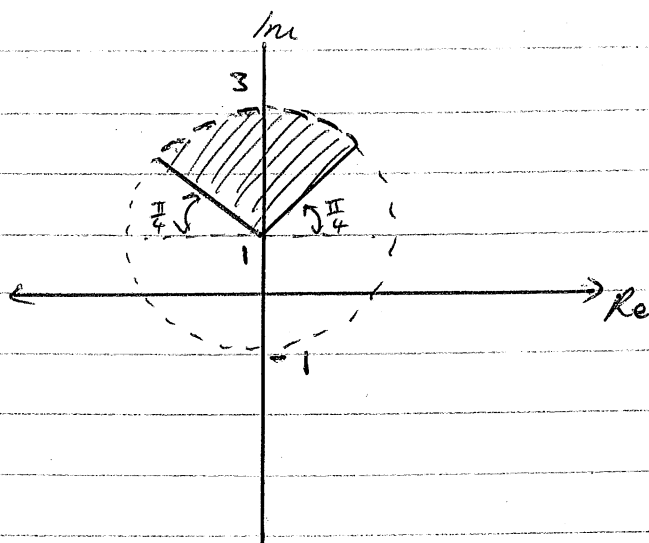
$$= 2^9 (\cos 6\pi + i \sin 6\pi)$$

$$= 2^9 (1 + i(0))$$

$$= 2^9$$

$$= 512$$

(b)



(c) let $a+ib = \sqrt{8-6i}$

$$a^2 - b^2 + 2abi = 8 - 6i$$

$$a^2 - b^2 = 8 \quad \text{--- (1)}$$

$$2ab = -6$$

$$ab = -3 \quad \text{--- (2)}$$

$$(a^2 + b^2)^2 = (a^2 - b^2)^2 - 4(ab)^2$$

$$= 64 + 4(-3)^2$$

$$= 100$$

$$a^2 + b^2 = 10 \quad \text{--- (3)}$$

$$\textcircled{1} + \textcircled{3}$$

$$2a^2 = 18$$

$$= \frac{\pi}{12}$$

$$(b) \int_0^1 \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$$

$$u^6 = x$$

$$\frac{dx}{du} = 6u^5$$

$$dx = 6u^5 du$$

$$\text{when } x=1, u=1$$

$$x=0, u=0$$

$$= \int_0^1 \frac{6u^5 du}{u^3 + u^2}$$

$$= 6 \int_0^1 \frac{u^3 du}{1+u}$$

$$= 6 \int_0^1 \frac{u^3+1}{1+u} du - 6 \int_0^1 \frac{du}{1+u}$$

$$= 6 \int_0^1 \frac{(u+1)(u^2-u+1)}{u+1} du - 6 \int_0^1 \frac{du}{u+1}$$

$$= 6 \left[\frac{u^3}{3} - \frac{u^2}{2} + u - \ln(u+1) \right]_0^1$$

$$= 6 \left[\frac{1}{3} - \frac{1}{2} + 1 - \ln 2 - 0 \right]$$

$$= 5 - 6 \ln 2$$

$$(c) I = \int e^x \cos x dx$$

$$u = \cos x \quad \begin{matrix} \swarrow v' = e^x \\ \searrow v = e^x \end{matrix}$$

$$u' = -\sin x$$

$$I = e^x \cos x + \int e^x \sin x dx + C$$

$$u = \sin x \quad \begin{matrix} \swarrow v' = e^x \\ \searrow v = e^x \end{matrix}$$

$$u' = \cos x$$

$$I = e^x \cos x + e^x \sin x - \int e^x \cos x dx + k$$

$$I = e^x (\cos x + \sin x) - I + k$$

$$2I = e^x (\cos x + \sin x) + k$$

$$I = \frac{e^x}{2} (\cos x + \sin x) + k, \text{ where } k \text{ is a constant.}$$

$$a^2 = 9$$

$$a = \pm 3$$

$$\text{when } a = 3$$

$$b = -1$$

$$a = -3$$

$$b = 1$$

$$3 - i, -3 + i$$

$$(d) \quad z = x + iy$$

$$\bar{z} = x - iy$$

$$z\bar{z} = x^2 + y^2$$

$$3(x^2 + y^2) + 2(2iy) = 39 + 12i$$

$$3(x^2 + y^2) = 39$$

$$x^2 + y^2 = 13$$

$$x^2 + (3)^2 = 13$$

$$x^2 = 4$$

$$x = \pm 2$$

$$4y = 12$$

$$y = 3$$

$$z = 2 + 3i, -2 + 3i$$

$$(e) \quad z = \cos \theta + i \sin \theta$$

$$1 + z = 1 + \cos \theta + i \sin \theta$$

$$= 2 \cos^2 \frac{\theta}{2} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} i$$

$$= 2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$$

$$\text{since } \cos \theta = 2 \cos^2 \frac{\theta}{2} - 1$$

$$\& \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

Question 2

$$(a) \quad \int_0^{\frac{1}{4}} \frac{dx}{\sqrt{1-4x^2}}$$

$$= \frac{1}{2} \int_0^{\frac{1}{4}} \frac{dx}{\sqrt{\frac{1}{4} - x^2}}, \quad a = \frac{1}{2}$$

$$= \frac{1}{2} \left[\sin^{-1} 2x \right]_0^{\frac{1}{4}}$$

$$= \frac{1}{2} \left[\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right]$$

$$= \frac{1}{2} \left(\frac{\pi}{6} \right)$$

$$(d) \int \frac{dx}{x^2 - x - 6}$$

$$= \int \frac{dx}{(x-3)(x+2)}$$

$$\frac{1}{(x-3)(x+2)} = \frac{a}{x-3} + \frac{b}{x+2}$$

$$1 = a(x+2) + b(x-3)$$

$$\text{let } x=3$$

$$1 = 5a$$

$$a = \frac{1}{5}$$

$$\text{let } x=-2$$

$$1 = -5b$$

$$b = -\frac{1}{5}$$

$$\therefore \int \frac{dx}{x^2 - x - 6} = \frac{1}{5} \int \frac{dx}{x-3} - \frac{1}{5} \int \frac{dx}{x+2}$$

$$= \frac{1}{5} \ln(x-3) - \frac{1}{5} \ln(x+2) + C$$

$$= \frac{1}{5} \ln \left(\frac{x-3}{x+2} \right) + C$$

$$(e) \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$$

$$= \int_0^1 \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= 2 \int_0^1 \frac{dt}{1+t^2+1-t^2+2t}$$

$$= 2 \int_0^1 \frac{dt}{2(1+t)}$$

$$= \int_0^1 \frac{dt}{1+t}$$

$$= \left[\ln(1+t) \right]_0^1$$

$$= \ln 2 - \ln 1 = \ln 2$$

$$t = \tan \frac{x}{2}$$

$$\frac{x}{2} = \tan^{-1} t$$

$$x = 2 \tan^{-1} t$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$\text{when } x = \frac{\pi}{2}, t = 1$$

$$x = 0, t = 0$$

QUESTION 3

(a) Given $x^3 + kx + 2 = 0$ (A) with roots α, β, γ .

$$(i) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{k}{-2} = \left(-\frac{1}{2}k\right) \checkmark$$

(ii) Let $x = x^2$

$\Rightarrow x = \pm\sqrt{x}$ in the above (A)

$$[NB \sum \alpha = +\frac{b}{a} = k]$$

$$[NB \alpha\beta\gamma = -\frac{c}{a} = -2]$$

$$\therefore (\pm\sqrt{x})^3 + k(\pm\sqrt{x}) + 2 = 0.$$

$$\pm x\sqrt{x} \pm k\sqrt{x} = -2.$$

$$\pm \sqrt{x}(x+k) = -2.$$

$$\therefore x(x+k)^2 = 4.$$

$$x(x^2 + 2kx + k^2) = 4.$$

$$\boxed{x^3 + 2kx^2 + k^2x - 4 = 0.} \checkmark \checkmark$$

$$(iii) \text{ From (A) } \alpha^3 + k\alpha + 2 = 0$$

$$\beta^3 + k\beta + 2 = 0$$

$$\checkmark \gamma^3 + k\gamma + 2 = 0$$

Adding we get:

$$\alpha^3 + \beta^3 + \gamma^3 + k(\alpha + \beta + \gamma) + 6 = 0.$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 + k \times 0 + 6 = 0$$

$$\boxed{\alpha^3 + \beta^3 + \gamma^3 = -6.} \checkmark \checkmark$$

(b) (i) Let $P(x) = (x-\alpha)^2 Q(x)$.

$$P'(x) = 2(x-\alpha) Q(x) + (x-\alpha)^2 Q'(x).$$

$$\text{ie } P'(x) = (x-\alpha)(2Q(x) - (x-\alpha)Q'(x)).$$

$$= (x-\alpha) R(x)$$

$$\therefore P'(\alpha) = (\alpha-\alpha) \cdot R(\alpha).$$

$$= 0 \cdot R(\alpha)$$

$$= 0.$$

✓✓

(ii). Let $P(x) = 18x^3 + 3x^2 - 28x + 12 = 0$

$$P'(x) = 54x^2 + 6x - 28 = 0$$

$$P'(x) = 2(27x^2 + 3x - 14) = 0$$

$$= 2(9x+7)(3x-2) = 0.$$

$$\therefore x = \frac{2}{3}, -\frac{7}{9}$$

Consider $x = \frac{2}{3}$.

$$P\left(\frac{2}{3}\right) = 18 \cdot \left(\frac{2}{3}\right)^3 + 3 \cdot \left(\frac{2}{3}\right)^2 - 28 \cdot \frac{2}{3} + 12.$$

$$= 18 \times \frac{8}{27} + 3 \times \frac{4}{9} - \frac{56}{3} + 12.$$

$$= \frac{16}{3} + \frac{4}{3} - \frac{56}{3} + 12.$$

$$= \frac{16+4-56+36}{3}$$

$$= 0.$$

✓✓✓

ie $P\left(\frac{2}{3}\right) = P'\left(\frac{2}{3}\right) = 0$

$\therefore$ Roots are $\frac{2}{3}, \frac{2}{3}$ and α .

where $\sum \text{roots} = \frac{2}{3} + \frac{2}{3} + \alpha = -\frac{3}{18} = -\frac{1}{6}$.

$$\therefore \frac{4}{3} + \alpha = -\frac{1}{6}$$

$$\alpha = -\frac{9}{6} = -\frac{3}{2}$$

$$(c) \quad f(x) = \frac{11-x}{x^2-x-2} = \frac{11-x}{(x-2)(x+1)}$$

$$\begin{aligned} f'(x) &= \frac{(x^2-x-2)(-1) - (11-x)(2x-1)}{(x-2)^2(x+1)^2} \\ &= \frac{-x^2+x+2 - (22x-11+2x^2-2x)}{(x-2)^2(x+1)^2} \\ &= \frac{-x^2+x+2 - (23x-11+2x^2)}{(x-2)^2(x+1)^2} \\ &= \frac{x^2-22x+13}{(x-2)^2(x+1)^2} \end{aligned}$$

For $f'(x) = 0$

$$\begin{aligned} x &= \frac{22 \pm \sqrt{484 - 52}}{2} \\ &= \frac{22 \pm \sqrt{432}}{2} \\ &= \frac{22 \pm 12\sqrt{3}}{2} \\ &= 11 \pm 6\sqrt{3} \\ &\approx 21.4, 0.6. \end{aligned}$$

$\therefore$ st. pts at $(0.6, -4.6)$ & $(21.4, -0.02)$

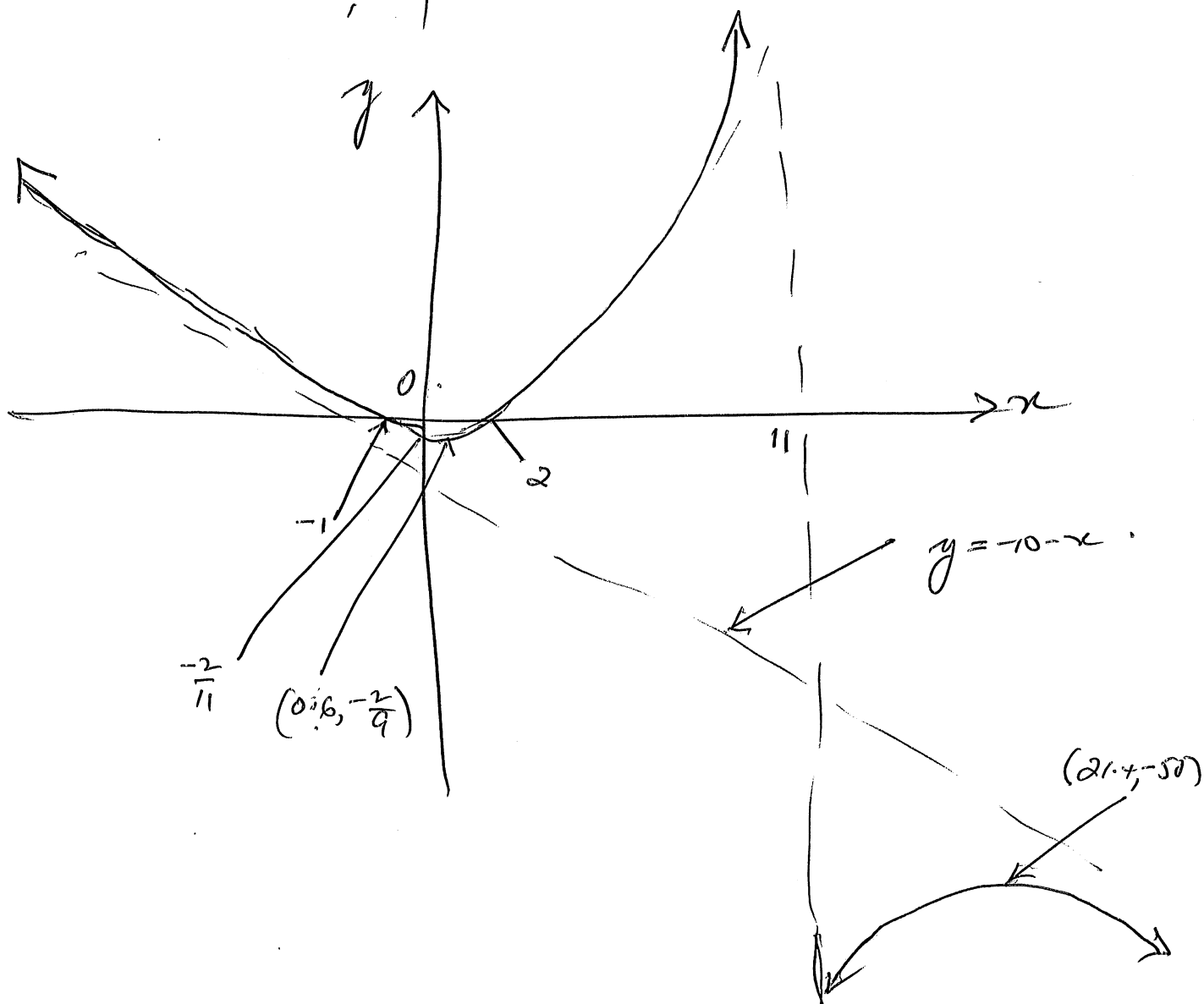
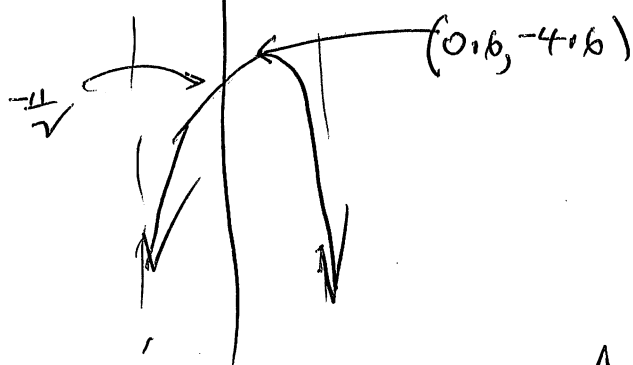
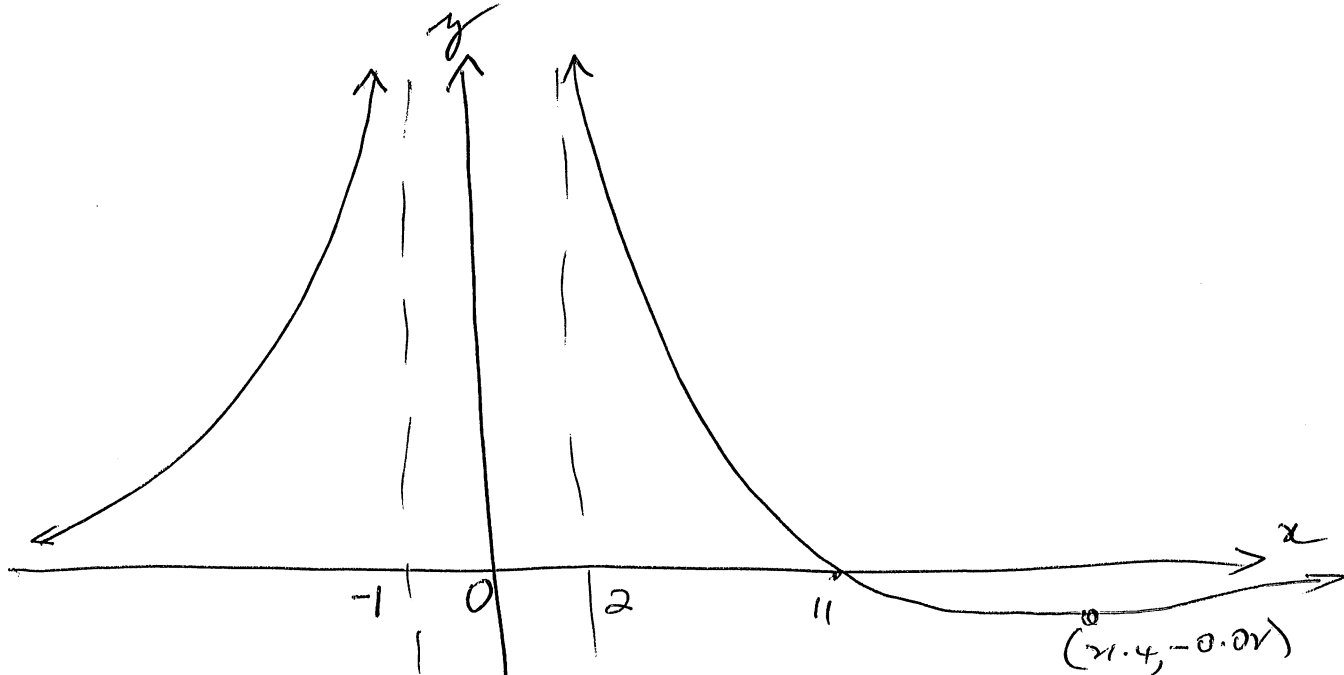
Testing

x	0	0.6	1
y'	$\frac{13}{4}$	0	-2

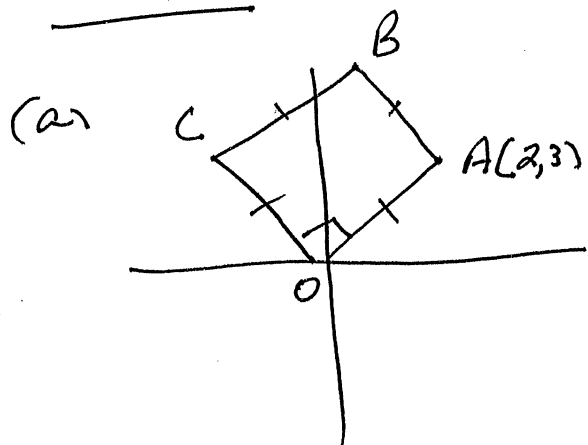
$\therefore$ REL. MAX
at $(0.6, -4.6)$

x	21	21.4	22
y'	-2.009 -4.6×10^{-5}	0	6.14×10^{-5}

REL MIN
at
 $(21.4, -0.02)$



QUESTION 4.



(i) C is given by.

$$i(2+3i) = \boxed{-3+2i} \quad \checkmark$$

(ii) B is given by.

$$-3+2i+2+3i = \boxed{-1+5i} \quad \checkmark$$

$$\begin{aligned} \text{(iii)} \quad A' &= \cos 45^\circ (2+3i) \\ &= \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}\right)(2+3i) \\ &= \frac{1}{\sqrt{2}}(1+i)(2+3i) \\ &= \frac{1}{\sqrt{2}}(-1+5i) \\ &= \boxed{-\frac{1}{\sqrt{2}} + \frac{5}{\sqrt{2}}i} \\ &\quad \text{OR } \left(-\frac{\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i\right) \end{aligned}$$

✓✓

$$\begin{aligned} B' &= \frac{1}{\sqrt{2}}(1+i)(-1+5i) \\ &= \frac{1}{\sqrt{2}}(-6+4i) \\ &= \boxed{-\frac{6}{\sqrt{2}} + \frac{4}{\sqrt{2}}i} \\ &\quad \text{OR } (-3\sqrt{2}+2\sqrt{2}i) \end{aligned}$$

$$\begin{aligned} C' &= \frac{1}{\sqrt{2}}(1+i)(-3+2i) \\ &= \frac{1}{\sqrt{2}}(-5-i) \\ &= \boxed{-\frac{5}{\sqrt{2}} - \frac{1}{\sqrt{2}}i} \\ &\quad \text{OR } \left(-\frac{5\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) \end{aligned}$$

(b)

$$\begin{aligned} \int_0^a f(x) dx &= \int_{a_0}^0 f(a-t) \cdot dt \quad \text{let } x=a-t. \\ &= -\int_{a_2} f(a-t) dt \\ &= \int_a^0 f(a-x) dx \\ &= \int_0^a f(a-x) dx. \end{aligned}$$

✓✓

$$\begin{aligned}
 (11) \quad \int_0^2 x^2 (2-x)^9 dx &= \int_0^2 (2-x)^2 \cdot x^9 dx \\
 &= \int_0^2 (4-4x+x^2) x^9 dx \\
 &= \int_0^2 (4x^9 - 4x^{10} + x^{11}) dx \\
 &= \left[\frac{4x^{10}}{10} - \frac{4x^{11}}{11} + \frac{x^{12}}{12} \right]_0^2 \\
 &= \frac{2}{5} \cdot 2^{10} - \frac{4}{11} 2^{11} + \frac{2^{12}}{12} \\
 &= 2^{10} \left[\frac{2}{5} - \frac{8}{11} + \frac{1}{3} \right] \\
 &= 2^{10} \times \frac{66 - 110 + 55}{165} \\
 &= 2^{10} \times \frac{11 - 110}{165} \\
 &= \frac{2^{10}}{165} \\
 &= \frac{1024}{165} \\
 &= 6 \frac{34}{165} \quad \checkmark \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (C) \quad (i) \quad \ddot{x} &= \frac{d}{dn} \left(\frac{1}{r} v^2 \right) \\
 &= \frac{d}{dn} \frac{1}{r} (36 - 4x^2) \\
 &= \frac{d}{dn} (18 - 2x^2) \\
 &= -4x
 \end{aligned}$$

$$(ii) \quad T = \frac{2\pi}{n}$$

where $n = v$

$$T = \frac{2\pi}{v}$$

$\therefore \ddot{x} \propto x \therefore$ S.H.M.

is of the form $\ddot{x} = -n^2 x$

$= \pi$ sec.

(d) Let $z^5 = 1$.

$(\cos \theta + i \sin \theta)^5 = 1$.

$\cos 5\theta + i \sin 5\theta = 1 + 0i$

$\therefore \cos 5\theta = 1$

$5\theta = 2k\pi, \quad k \in \mathbb{Z}.$

$\theta = \frac{2k\pi}{5}$.

$= 0, \pm \frac{2\pi}{5}, \pm \frac{4\pi}{5}.$

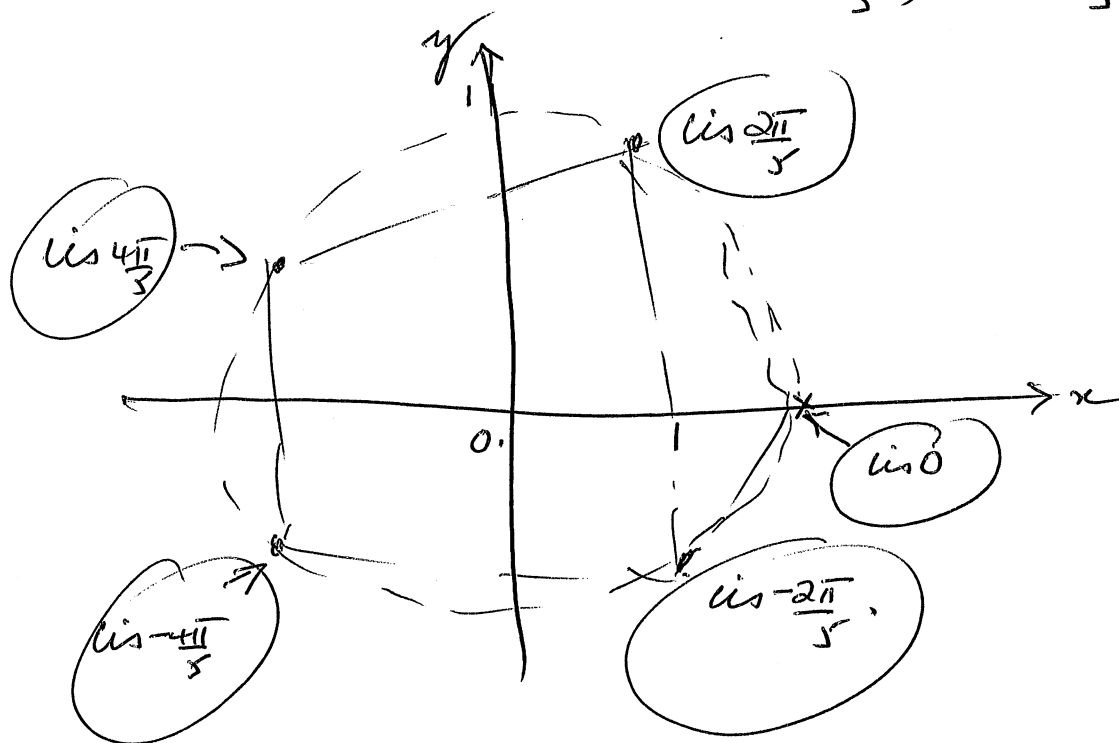
$\therefore$ roots are $\cos 0, \cos \pm \frac{2\pi}{5}, \cos \pm \frac{4\pi}{5}.$

[where $z = r(\cos \theta + i \sin \theta)$.

now $|z| = r$.

$|z^5| = |z|^5 = 1$

$\therefore |z| = r = 1.$



SECTION C

QUESTIONS

(a) $z^2 + (1+2i)z - (2-4i) = 0$

$$z = \frac{-(1+2i) \pm \sqrt{5-12i}}{2}$$

let $5-12i = (a+ib)^2$

$$5-12i = (a^2-b^2) + 2abi$$

$$a^4 - 5a^2 - 36 = 0$$

$$(a^2-9)(a^2+4) = 0$$

$$a = \pm 3 \Rightarrow b = \mp 2$$

(3)

$$\therefore z = \frac{-(1+2i) \pm (\pm 3 \mp 2i)}{2}$$

$$\Rightarrow z = 1-2i \text{ or } z = -2$$

(b) $I_n = \int_0^{\pi/4} \tan^n x \, dx$

(i) $\therefore I_1 = \int_0^{\pi/4} \tan x \, dx$

$$= [\ln(\sec x)]_0^{\pi/4} \quad (1)$$

$$= \ln \sqrt{2} \text{ or } \frac{1}{2} \ln 2$$

(ii) $I_n = \int_0^{\pi/4} \tan^n x \, dx \quad (2)$

$$= \int_0^{\pi/4} \tan^{n-2} x \cdot \tan^2 x \, dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx - \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$I_n = \frac{1}{n-1} \left[\tan^{n-1} x \right]_0^{\pi/4} - [I_{n-2}]$$

$$\therefore I_n = \frac{1}{n-1} - I_{n-2}$$

$$\Rightarrow I_n + I_{n-2} = \frac{1}{n-1}$$

(iii) $J = \int_0^{\pi/4} \tan^n x \, dx$

$$I_n = \frac{1}{n-1} - I_{n-2}$$

$$\therefore I_{11} = \frac{1}{10} - I_9$$

$$= \frac{1}{10} - \left[\frac{1}{8} - I_7 \right]$$

$$= \frac{1}{10} - \frac{1}{8} + \frac{1}{6} - I_5 \quad (2)$$

$$= \frac{1}{10} - \frac{1}{8} + \frac{1}{6} - \left[\frac{1}{4} - I_3 \right]$$

$$= \frac{1}{10} - \frac{1}{8} + \frac{1}{6} - \frac{1}{4} + \frac{1}{2} - I_1$$

$$= \frac{47}{120} - \frac{1}{2} \ln 2$$

(d) Since real pol, if $2+i$ is a root $\Rightarrow 2-i$ also a root

$$\text{Product of roots} = -15$$

$$\therefore (2-i)(2+i)x = -15$$

$$5x = -15 \quad (4)$$

$$x = -3$$

Roots $2+i, 2-i, -3$

(c) $\cos 4\theta + i \sin 4\theta = (c+is)^4$

$$\cos 4\theta = \text{Re} [(c+is)^4] \quad (2)$$

$$= \text{Re} [c^4 + 4c^3si - 6c^2s^2 - 4cs^3i + s^4]$$

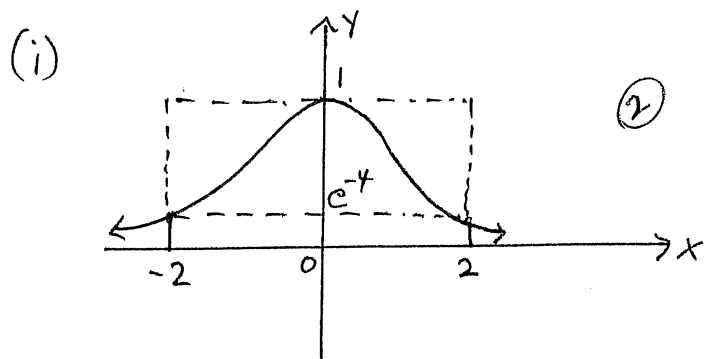
$$= c^4 - 6c^2s^2 + s^4$$

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta + 6\cos^4 \theta + (1-c^2)^2$$

$$\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$$

QUESTION 6

(a) $y = e^{-x^2}$



(ii)

Area small Rect $< \int_{-2}^2 e^{-x^2} dx < \text{Area large Rect}$

$$4(e)^{-4} < \int_{-2}^2 e^{-x^2} dx < 4(1)$$

$$\Rightarrow 1 < \int_{-2}^2 e^{-x^2} dx < 4$$

[NOTE: $4e^{-1} > 1$]

(iii)

-2 -1 0 1 2

$$\int_{-2}^2 e^{-x^2} dx \approx \frac{1}{3} \left[f(-2) + 4f(-1) + f(0) \right] + \frac{1}{3} \left[f(0) + 4f(1) + f(2) \right]$$

ie ≈ 1.66

(b) (i) $x = Vt \cos \theta$ ① $y = Vt \sin \theta - \frac{gt^2}{2}$

(ii) Particle at P $\Rightarrow x = -y$

Since OM = MP ①

$$\Rightarrow Vt \cos \theta = \frac{gt^2}{2} - Vt \sin \theta$$

$$\therefore t = \frac{2V(\sin \theta + \cos \theta)}{g}$$

(iii) OM = x = Vt \cos \theta

$$= V \cos \theta \cdot \left[\frac{2V(\sin \theta + \cos \theta)}{g} \right]$$

$$= \frac{V^2}{g} \left[2(\sin \theta + \cos \theta) \cos \theta \right]$$

② $= \frac{V^2}{g} \left[\sin 2\theta + 2 \cos^2 \theta \right]$

$$= \frac{V^2}{g} \left[\sin 2\theta + \cos 2\theta + 1 \right]$$

(iv) Range $r = \frac{V^2}{g} \sin 2\theta$

OM = $\frac{4}{3}r = \frac{V^2}{g} (\sin 2\theta + \cos 2\theta + 1)$ ②

$$\Rightarrow \frac{4}{3} \left(\frac{V^2}{g} \sin 2\theta \right) = \frac{V^2}{g} (\sin 2\theta + \cos 2\theta + 1)$$

$$\therefore 4 \sin 2\theta = 3 \sin 2\theta + 3 \cos 2\theta + 3$$

ie $\boxed{\sin 2\theta - 3 \cos 2\theta = 3}$

Now $2 \sin \theta \cos \theta = 3(1 + \cos 2\theta)$

$$2 \sin \theta \cos \theta = 3(2 \cos^2 \theta)$$

$(\div \cos^2 \theta)$

$$\Rightarrow \frac{2 \sin \theta}{\cos \theta} = 6$$

$\therefore \boxed{\tan \theta = 3}$ ②